

phase_06_GAP_AUDIT

Phase 6 — Gap Audit (Multi-Cluster Federation)

Field	Value
Auditor	Engineering auditor (code-grounded)
Date	2026-06-01
Scope	docs/research/ PHASE_6_ROADMAP.md deliverables vs. actual code in pkg/, internal/, cmd/, api/, security/ submodule
Honest completion	~4% of Phase 6 deliverables DONE

3-Axis Package Status (Refreshed 2026-06-04, HXC-939)

Why this section exists: “exists” ≠ “used”. wired is **measured** via `go list -deps ./cmd/... | grep -Fx <module-path>/<pkg>` (module = `github.com/HelixDevelopment/helix_cluster`), not assumed. **“Completed (registry) ≠ wired”** — Completed only means source+tests exist; it does NOT prove a shipped binary reaches it. All 11 planned federation packages are **unimplemented**; only single-cluster base packages exist.

Package	implemented	wired (reachable from cmd/)	tested
pkg/federation	NO	n/a	n/a
pkg/swim/hierarchical	NO	n/a	n/a
pkg/crdt (+ pkg/crdt/merkle)	NO	n/a	n/a
pkg/hlc	NO	n/a	n/a
pkg/nat traversal	NO	n/a	n/a

Package	implemented	wired (reachable from cmd/)	tested
pkg/cilium	NO	n/a	n/a
pkg/spiffe/ federation	NO	n/a	n/a
pkg/gitops	NO	n/a	n/a
internal/ cell / internal/ federation / internal/ chaos	NO	n/a	n/a
cmd/helix- federation	NO	n/a	n/a
pkg/swim (base, would be <i>extended</i> for Phi- accrual/ gateway-relay)	yes (base only)	yes	yes
pkg/ wireguard (base, would be <i>extended</i> for multi-cell)	yes (base only)	yes	yes
pkg/security (base, would be <i>extended</i> for SPIFFE federation)	yes (base only)	yes	yes
pkg/ scheduler (base, would be <i>extended</i> for inter-cell)	yes (base only)	yes	yes

Note: the four base packages are implemented+wired+tested for single-cluster Phase 2–5 scope but carry **no** federation behaviour, so they are not Phase 6 deliverables. No Phase 6 package is orphaned because no Phase 6 package exists yet.

One-line summary: Phase 6 (federation: cells, hierarchical SWIM, CRDT/HLC, NAT traversal, SPIFFE federation, Cilium/Karmada/ArgoCD) is **essentially not started** — all 11 planned packages are absent and there are zero federation/cross-cell terms in the Go codebase; the only Phase-6-adjacent code is single-cluster scaffolding (pkg/swim, pkg/wireguard) plus a documented NAT-traversal **stub** that returns “not implemented”.

Method / Anti-bluff notes

- A deliverable is **DONE** only if implemented AND covered by real-behavior tests. Stub/TODO/mock-only/error-path-only ⇒ PARTIAL or MISSING.
- Searched the full Go tree for federation vocabulary (karmada, propagationpolicy, merkle, hybrid logical, g-counter, lww-register, cluster mesh, applicationset, cross-cell, inter-cell): **0 matches**.
- Searched for the 11 planned package dirs (federation, crdt, hlc, nattraversal, cilium, gitops, internal/cell, internal/federation, internal/chaos, pkg/swim/hierarchical, pkg/spiffe/federation): **none exist** (pkg/testing/chaos is unit-test chaos helpers, unrelated).
- Recent hardening (cost-aware scheduling, pkg/fiber, pkg/gpuattest, e2ee/attestation in the security submodule, etcd/NATS/Postgres integration tests) is real but is **single-cluster Phase 5/8 scope**, not Phase 6 federation. Credited only where the roadmap names it an “existing package to extend”.

Deliverable Status Table

Deliverable (roadmap)	Status	Evidence file:line	Notes
pkg/federation — cell registry, lifecycle state machine, federated API proxy	MISSING	no dir under pkg/	No package; no Cell/join/evacua anywhere.
pkg/swim/hierarchical — two-tier LAN+WAN delegate gossip	MISSING	pkg/swim/ (no hierarchical/)	pkg/swim is single-tier only; no del gateway/WAN code.
pkg/crdt — G-Counter, LWW-Register, OR-Set + Merkle delta	MISSING	no dir under pkg/	Zero CRDT/Merkle code in tree.
pkg/hlc — Hybrid Logical Clock	MISSING	no dir under pkg/	pkg/swim/clock.go is a plain mor helper, not an HLC.
pkg/nattraversal — STUN/TURN/ICE + UDP hole punch + QUIC	MISSING (stub adjacent)	pkg/wireguard/nat_traversal.go:16–41	DiscoverExternalAddress hits ip the http client returns “not available build”; SetupPortMapping/RemovePortMapping return "not implemented". Tests assert the error only (pkg/wireguard/package_test.go:288–308). No S TURN/ICE/QUIC.
pkg/cilium — Cluster Mesh client	MISSING	no dir under pkg/	No Cilium/eBPF integration.

Deliverable (roadmap)	Status	Evidence file:line	Notes
pkg/spiffe/federation — trust-bundle exchange across cells	MISSING	pkg/security/spiffe.go (no federation refs)	pkg/security/spiffe.go exists for single-domain SPIFFE; no trust-bundle exchange or federation code.
pkg/gitops — ArgoCD ApplicationSet federation	MISSING	no dir under pkg/	No ArgoCD/ApplicationSet code.
internal/cell — gateway mgmt, inter-cell WG wiring, SWIM delegate config	MISSING	no dir under internal/	Absent.
internal/federation — Karmada/OCM, PropagationPolicy, kubectl aggregation	MISSING	no dir under internal/	Absent.
internal/chaos — 12 chaos experiments (CE-01..CE-12)	MISSING	no dir under internal/	Absent (pkg/testing/chaos is under test, no fault helpers).
cmd/helix-federation binary	MISSING	cmd/ (not present)	No federation CLI; no helixctl federation / cell join.
Ext: pkg/swim Phi-accrual + gateway-relay suspicion	PARTIAL	pkg/swim/failure_detector.go:8–33	FailureDetector uses a fixed timeout for timer suspicion, NOT Phi-accrual. No gateway-relay path. Base SWIM (prober/suspicion) is real and tested for single-cluster.
Ext: pkg/wireguard multi-cell peers + zero-downtime key rotation	PARTIAL	pkg/wireguard/keyrotation.go:97–141	RotateKeysTracked rotates and reuses the <i>previous</i> key for audit and is tested (keyrotation_config_test.go:60–65) but there is no zero-downtime overlap window (“add new before remove old” 100ms window) and no multi-cell/inter-cell route advertising. Single-cluster mechanism.
Ext: pkg/security SPIFFE/SPIRE federation + OPA cross-cluster	PARTIAL	pkg/security/spiffe.go, internal/policy/	Single-domain SPIFFE + single-cluster SPIRE policy exist; cross-cluster trust bundle and OPA admission absent.
Ext: pkg/scheduler inter-cell Karmada	PARTIAL	pkg/scheduler/cost_gpu.go:1–30	Real single-cluster cost-aware GPU placement plugin (tested), but no inter-cell Karmada propagation; “cost reduction” ≠ cross-cell bursting.

Deliverable (roadmap)	Status	Evidence file:line	Notes
propagation + spot scoring			
Ext: internal/ node cell agent / federation identity attestation	MISSING	internal/node/node.go	Node agent has no cell/federation/ti selection logic.
Exit gate: real multi-cell integration tests (CLAUDE-1)	MISSING	n/a	No multi-cell testbed; the only WG asserts the stub error path.
Exit gate: e2e operator flows (cell join, -- all-cells, GitOps)	MISSING	n/a	No CLI surface exists to exercise.

TOP IMPLEMENTABLE GAPS (Go, no new infra)

Prioritized for maximum federation value while remaining pure-Go and locally testable. Each is mutation-pairable per §1.1.

1. pkg/hlc — Hybrid Logical Clock (P0, foundational)

Target dir: pkg/hlc/ **Spec:** Implement a Hybrid Logical Clock: a Clock holding {wallMillis, logical} with Now() (advances against time.Now), Update(remote Timestamp) (max of local/remote wall, bumps logical on ties, clamps drift to a configurable bound e.g. 10 ms→error), and a total-order Timestamp.Compare. Pure-Go, deterministic with an injectable wall-clock func. This is the timestamp substrate every CRDT type needs and unblocks pkg/crdt.

Acceptance test (mutation-pairable): Feed an out-of-order remote timestamp newer than local wall; assert returned timestamp is strictly greater than both inputs and that two events at identical wall time get distinct logical counters preserving causal order. Mutant: change Update to ignore the remote logical (use only wall) → causal-order assertion must fail.

2. pkg/crdt — G-Counter, LWW-Register, OR-Set (P0)

Target dir: pkg/crdt/ **Spec:** Implement three state-based CRDTs over pkg/hlc timestamps with Merge(other) that is commutative, associative, idempotent. G-Counter (per-replica counter map, value=sum); LWW-Register (value + HLC, last-writer-wins on Compare); OR-Set (add/remove with unique tags). No network — just the convergence algebra. **Acceptance test:** Property test merging the same set of ops in randomized orders across 3 replicas; assert all replicas converge to byte-identical state and that Merge is idempotent (merging twice = once). Mutant: make OR-Set Remove delete by element instead of by tag → concurrent add/remove convergence test must fail.

3. pkg/federation — cell registry + lifecycle state machine (P0)

Target dir: pkg/federation/ **Spec:** Pure-Go cell model: Cell{ID, Gateways, State} and an explicit FSM with transitions Joining→Syncing→Active→Evacuating→Detached (plus Failed). Provide a thread-safe Registry (add/remove/list cells, snapshot) and a Transition(cell, event) that rejects illegal transitions. No real networking — the lifecycle/validation logic is the deliverable; wire to transport later. **Acceptance test:** Drive a cell through the legal happy path and assert each state; then attempt an illegal jump (Detached→Active) and assert a typed error + unchanged state. Mutant: remove the illegal-transition guard → the rejection test must fail.

4. Phi-accrual failure detector in pkg/swim (P1)

Target dir: pkg/swim/ (new phi.go, used by failure_detector.go) **Spec:** Add a Phi-accrual detector: a sliding window of inter-arrival heartbeat intervals producing a phi(now) suspicion value from the sample mean/variance; expose SuspicionLevel(memberID) and a threshold-based IsSuspect. Replace/augment the current fixed-timeout FailureDetector so suspicion adapts to observed latency. Pure-Go, injectable clock. **Acceptance test:** Feed a steady 1 s heartbeat stream then stop; assert phi stays low while heartbeats arrive and crosses threshold within a bounded time after they stop; assert a jittery-but-alive stream does NOT cross threshold. Mutant: drop the variance term (use mean only) → the jitter-tolerance assertion must fail.

5. pkg/crdt Merkle anti-entropy delta sync (P1, depends on #2)

Target dir: pkg/crdt/merkle/ **Spec:** Build a Merkle tree over a keyspace of CRDT registers; Diff(remoteRoot) returns only the divergent key set so reconciliation ships deltas, not full state (the roadmap's <5 KB/s gossip budget). Pure-Go over an in-memory map; transport-agnostic. **Acceptance test:** Two trees differing in 1 of N keys; assert Diff returns exactly that key and that syncing it makes roots equal. Mutant: hash only leaf values, not subtree children → divergence in a deep key must still be detected, so a “single changed deep key” test must fail under the mutant.

6. pkg/federation federated API aggregation model (P1, depends on #3)

Target dir: pkg/federation/ (new aggregate.go) **Spec:** Pure-Go fan-out/merge: given a set of Active cells from the Registry and a per-cell query func (interface, mocked in unit tests), aggregate results with partial-failure semantics (return successes + a per-cell error map; never fail-closed on one cell). This is the engine behind helixctl get ... --all-cells. **Acceptance test:** Aggregate over 3 fake cells where one returns an error; assert 2 results returned + 1 recorded error and no panic. Mutant: make aggregation abort on first error → partial-success assertion must fail.

DEFERRED (need external infra / hardware / non-Go)

Item	Reason deferred
pkg/nat traversal real STUN/TURN/ICE + UDP hole punch + QUIC	Requires real NAT devices / STUN-TURN servers and a network test matrix (4 NAT types) to validate honestly; not unit-testable to a real-behavior standard locally.
pkg/cilium Cluster Mesh data plane	Requires Cilium + eBPF + a Kubernetes multi-cluster; out of pure-Go scope.
internal/federation Karmada/OCM PropagationPolicy	Requires running Karmada/OCM control planes and real K8s clusters.
pkg/gitops ArgoCD ApplicationSet federation	Requires a live ArgoCD instance + Git backend to prove end-user flow.
pkg/spiffe/federation trust-bundle exchange	Needs ≥ 2 SPIRE servers (nested trust domains) for real cross-cell SVID/bundle validation.
internal/chaos CE-01..CE-12 + Chaos Mesh	Requires Chaos Mesh + a multi-node testbed and partition injection; the 12 KPIs are infra-bound.
WireGuard zero-downtime key rotation overlap window	Kernel WireGuard interface + multi-host mesh needed to prove the 200 ms overlap doesn't flap (skipped on darwin today).
Velero DR / Prometheus federation / split-brain alerts	External backup + monitoring stacks; KPI measurement (RPO, alert latency) requires the running federation.